Teaching Technical Skills

From R. Martens and R.S. Vealey, *Successful Coaching*, 5th ed. (Champaign, IL: Human Kinetics, 2024), 156-158; 161-164.

Steps to Teach Technical Skills

Now we'll let you return to being the coach. Imagine that you begin your practice session with a modified game to emphasize the further development of a technical skill, but as the game progresses, you see a pattern of errors that indicates the need to review the fundamentals. You stop the game and teach the technique using the four steps shown in figure 10.2.

Let's learn more about each of these steps and describe how they are done using the games approach to coaching.

Step 1: Introduce the Technical Skill

Introduce the technical skill with enthusiasm expressed in actions and words. Speak clearly and use language your athletes can understand; the younger the athletes are, the simpler your words need to be. Be brief, too. Say what you have to say in less than three minutes. Avoid sarcasm, annoying mannerisms, and abusive language; they create a negative learning environment.

A good introduction involves

1. getting the team's attention,
2. arranging the team so everyone can see and hear you, and
3. naming the technique and explaining how it is used in the game.

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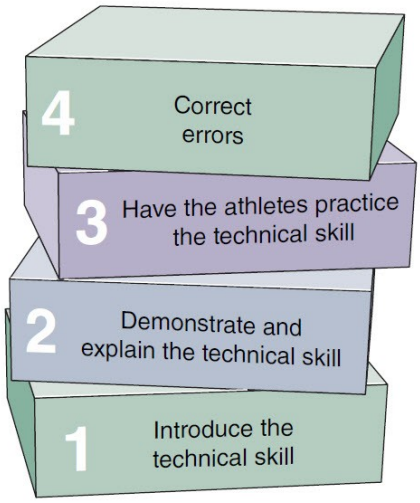


FIGURE 10.2 Four steps of teaching technical skills.

Figure 10.2 — Text in diagram

1. Introduce the technical skill
2. Demonstrate and explain the technical skill
3. Have the athletes practice the technical skill
4. Correct errors

Get the Team's Attention

Develop a regular routine in practice for starting each teaching session. Go to your usual place to begin the session and give a signal, such as blowing a whistle, to get your athletes' attention and bring them into the area. Position yourself to face the team when you speak to them and make direct eye contact with them to help keep their attention. Occasionally ask your players to repeat key points to maintain their involvement.

If a few athletes are inattentive, look directly at them, move closer to them, and politely but firmly ask for their attention. If this fails, have them move to where they cannot disrupt the session. Speak with these athletes either at an opportune time later during the practice or afterward.

Arrange the Team So Everyone Can See and Hear You

When you speak to your athletes, be sure to organize them so they can see and hear you. If they are milling around or crowding together, you will have more difficulty keeping their attention. Be certain that the background behind you is free of visual distractions and that athletes are not facing the sun. Try also to select a practice area with minimum noise so that athletes can hear you.

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Name the Technique and Explain How It Is Used

Naming technical skills is important so that you can make quick reference to them. If a skill is widely known by a certain name, use that one, such as the jump stop layup in basketball or wiping the block in volleyball. If not, select a short, descriptive title that is easy to remember.

Sometimes the reason for learning a technique is not obvious, especially to athletes with little experience. The better your athletes understand why they are learning a particular technique and how it fits into the total plan for playing the sport, the easier it is for them to develop a mental plan for learning it. Understanding why they are learning also increases their motivation to learn. Use popular examples to grab their attention: "Did you know that Rafael Nadal works on his footwork every day and credits that for his success? We're going to learn the footwork and positioning so you can forehand volley like Rafa."

Step 2: Demonstrate and Explain the Technical Skill

Demonstration and explanation are the primary ways to help your athletes acquire a mental plan for a technique. Someone who can perform the technique proficiently and whom the athletes respect should do the demonstration. If you cannot demonstrate a particular technique, you have several alternatives:

- Practice the technique until you are able to demonstrate it correctly.
- Ask someone who is skilled to demonstrate, perhaps a team member, an assistant coach, or a friend.
- Show a video to demonstrate the technique. You can find videos of many sport techniques on YouTube.

If none of these alternatives is possible and you cannot give an adequate demonstration, seriously reconsider teaching this technical skill, especially if there is any risk of injury in learning the technique and you don't know how to control this risk.

An effective demonstration and explanation consists of three steps:

1. Demonstrating and explaining
2. Relating the technique to previously learned techniques
3. Checking for understanding

Demonstrate and Explain

Follow these guidelines for giving demonstrations:

- Tell your athletes how the demonstration will be given and what to look for.
- Demonstrate the whole technique just as it would be performed in a competitive situation.

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- Demonstrate several times, showing how to perform the technique from different angles.
- If the technique is performed from a dominant side, demonstrate it for both left-handers and right-handers.
- If the technique is complex, demonstrate the major parts separately.
- If the technique is performed rapidly, demonstrate it at a slower speed so athletes can see the sequence of movements clearly.

During the demonstration, also explain how the technical skill is done. However, remember that it is harder for most athletes to convert words into a mental plan for performing the technique than it is to interpret the demonstration. Follow these guidelines for your explanations:

- Before the demonstration, point out one or two important things to which the athletes should attend.
- Keep your explanations simple and brief.
- Ensure that the explanation matches what is being demonstrated.
- Time your explanation either to prepare the athletes for what they will see or to reinforce what they just saw.



Keep demonstrations brief; tell athletes one or two things they should watch for in your demonstration.

Jonathan Newton / The Washington Post via Getty Images

Relate the Technique to Previously Learned Techniques

After the technique is initially demonstrated, relate it to any previously learned techniques. Why is this important? As you learned earlier, the motor programs for a technique are generalized rules. Thus, it is possible to transfer some of these rules for movement to the new technique being learned. For example, if you are teaching the

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tennis serve to beginners, tell and show your athletes how this skill is similar to throwing a ball.

Check for Understanding

Now check to see if your athletes understand how to perform the technique by inviting or asking questions. If necessary, repeat a question so everyone can hear. Keep your answers short and relevant. Don't begin explaining all the nuances of the technique at this point.

Step 3: Have the Athletes Practice the Technical Skill

Athletes should begin practicing the technique as soon as possible following the demonstration and explanation. This brings you to a critical decision: Should your athletes practice the whole technique, or should you break it into parts? See the box Whole Versus Parts Method of Practice to learn more.

Better Practice Principles

Once you have decided whether you will practice using the whole or the whole-parts-whole method, you're ready to plan your practice. Use the list of seven principles featured in the box for better technical skill practice based on the games approach. Now let's look at each of these principles in greater depth.

Seven Principles for Technical Skill Practice

1. Have athletes practice the right technique.
2. Have athletes practice the technique in gamelike conditions as soon as they can.
3. Keep practices short and frequent when teaching new techniques.
4. Use practice time efficiently.
5. Make optimal use of facilities and equipment.
6. Maximize learning and success using progressions, extensions, and refinements.
7. Make practice enjoyable.

Remember, too, that formal practice often doesn't provide enough time to master technical skills thoroughly. Teach your athletes to practice the techniques on their own if they can do so safely.

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Principle 1: Practice the Right Technique

One of the most common mistakes in designing practice experiences is having athletes perform drills that do not help them learn the techniques required to play the sport. A drill often teaches only the skill unique to that drill. Many coaches use drills merely because their own coaches used them in the past. Carefully analyze the drills you use. Select only those that you are confident will help your athletes learn the techniques needed to play the sport or else design your own. Practice like you play, and your athletes are more likely to play like they practice.

Principle 2: Practice the Technique in Gamelike Conditions

This principle is closely related to the first principle. The purpose of many drills is to limit the variety of choices to be made and responses to be performed. That's useful when athletes are initially learning complex techniques, but when such drills are overused and competitive simulations are few, athletes are not prepared to make choices and responses in the rapidly changing conditions of a game. Thus, as you learned from principle one, have your athletes practice what is relevant and pertinent when actually playing the sport.

To apply this second principle, athletes should also practice the technique at the speed it is to be performed in competition, provided it can be executed safely and with a reasonable degree of accuracy. This produces more rapid and effective learning than does emphasizing slow, accurate movements and gradually increasing the speed. On the other hand, if the technique requires both speed and accuracy, practice should give equal emphasis to both.

When you adopt the games approach to coaching, you'll be much more likely to apply the first two principles—practicing the right techniques in gamelike conditions.

Principle 3: Keep Practices Short and Frequent When Teaching New Techniques

When first learning a technique, athletes are likely to make many mistakes and tire quickly. Therefore, they should practice the technique frequently, but not for too long. In other words, when athletes must use considerable mental and physical effort to perform a technique, practice should be interspersed with either rest intervals or practice of another technique that uses different muscle groups and demands less effort.

Principle 4: Use Practice Time Efficiently

Table 10.1 lists some big practice time wasters and suggestions for improving your use of time. Identify the time wasters in your practices and find ways to make your practices more efficient.

Principle 5: Make Optimal Use of Facilities and Equipment

Design practice activities to make efficient use of your facilities, equipment, and assistant coaches. Consider not only maximum use but also best use.

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TABLE 10.1 Making Practices More Efficient

Time wasters	Time savers
Drills in which most of the athletes' time is spent waiting	Reorganize drills so athletes are more active; think about the maximum reps you can provide all athletes.
The coach talks too much	Keep demonstrations, explanations, and feedback concise.
Moving between activities in the practice schedule	Use a detailed practice plan for all activities and implement routines for athletes to follow when changing activities (and make this quick and efficient).
Dealing with athletes' misbehavior	Separate the misbehaving athlete from the team, have the team continue practicing, and then speak with the misbehaving athlete.
Insufficient or ill-prepared facilities or equipment	Set up practice facilities ahead of time and attend to details in this setup and organization.



Simulate gamelike conditions during practice sessions.

Derek Davis/Portland Press Herald/Getty Images

Principle 6: Maximize Learning and Success With Progressions, Extensions, Refinements

Practices should be challenging, yet provide some successful experiences for athletes. An important way to build success into every practice is to use appropriate teaching

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progressions for learning technical skills. This means that the coach designs the learning experience to proceed from simple to more complex tasks. For example, a teaching progression in learning to field a ground ball in softball would be (1) going through the motions of fielding a grounder without a ball; (2) fielding a slow-rolling grounder thrown by the coach; (3) fielding a faster-moving ground ball thrown by the coach; (4) fielding a slow-moving grounder hit off a bat; and (5) fielding a fast-moving grounder hit off a bat. Every skill should be taught with an appropriate progression that doesn't move too slow or too fast.

Another way to optimally challenge athletes in teaching skills is to use extensions, which means you extend the task to become more or less difficult depending on the learner. For example, to extend and make kicking a soccer ball into a goal more difficult, a defensive player is added. To extend and make kicking the ball into the goal less difficult, there is no defense.

A refinement is used to focus the learner on improving a specific aspect of the task. A refinement for adding to a learner's skill on a pick and roll in basketball would be to show her how to pin the defense with her backside as she rolls. Refinements are taught when athletes are ready to refine or improve on the complexity of the skill.

For athletes who learn quickly, add a refinement or extend the task so that it is a little more difficult and challenge them to improve. For an athlete who is struggling, change the practice so it is less difficult until the skill is learned. It's fine to have different athletes practicing different skills at different levels of difficulty.

Principle 7: Make Practice Enjoyable

You can avoid boring practice sessions by using a lot of variations on a specific technical theme in practice. Practice games, challenges, and gimmicks to add interest are helpful. Changing your practice schedule occasionally, being enthusiastic, and letting the team help plan practices also are useful ways to make practices more enjoyable for athletes.

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Making practices enjoyable is key to keeping young athletes in sport.

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Step 4: Correct Errors

Practice alone is not enough to learn a technique correctly. For practice to be productive, you must provide your athletes with two types of information to correct errors:

1. How the completed performance compared with the desired performance
2. How to change an incorrect performance to more closely approximate the desired performance

Both types of information are called *feedback*.

Observe and Evaluate Performance

The process of helping your athletes correct errors begins with you observing and evaluating their performances to determine the cause of the errors. Did they not learn the technical skill well, or do they need further development of tactical, physical, mental, or communication skills? This is one of the big challenges in coaching—identifying the correct cause of inadequate performances. If athletes consistently demonstrate good technique in practice, but not in games, then they probably need help with their mental skills. If they demonstrate good technique in practice but display poor judgment in using those skills in games, then you will want to emphasize tactical skill training.

"You can observe a lot just by watching."

YOGI BERRA

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There is no substitute for knowledge and experience in correcting technical errors. The better you understand a technical skill—not only how it is done correctly but what causes athletes to err—the more helpful you will be in correcting mistakes. Experience is the most common way to learn to correct errors, but you can expedite the slow process of learning by experience through the study of sport biomechanics and the technical skills in your sport. The use of video to observe your athletes more carefully can also be of great help.

One of the most common coaching mistakes is to provide inaccurate feedback and advice on how to correct errors. Don't rush into error correction; wrong feedback or poor advice will hurt the learning process more than no feedback or advice. If you are uncertain about the cause of a problem or how to correct it, continue to observe and analyze until you are more certain. As a rule, you should observe the error several times before attempting to correct it.

Use Video

You can use video for introducing technical and tactical skills and providing feedback to your athletes. Many videos are available for demonstrating most sport skills, and it is relatively inexpensive to purchase a video camera to tape practices and contests. In addition, companies like Dartfish (www.dartfish.com/) and Hudl Sportscode (www.hudl.com/products/sportscode) offer powerful video software to help you analyze technical skills and provide visual feedback to your athletes.

A Positive Approach to Correcting Errors

Throughout this book, we have encouraged you to use the positive approach in coaching your athletes. That is especially applicable when correcting errors. Staying positive when your athletes repeatedly perform a technical skill incorrectly or lack enthusiasm for learning can be a real challenge. It can certainly be frustrating to coach athletes who seemingly don't heed your advice and continue to make the same mistake. When an athlete doesn't seem to care, you may wonder why you should. It is perfectly normal to get frustrated at times when teaching skills. Nevertheless, part of successful coaching is controlling this frustration and continuing to search for a way to help your athletes.

A positive approach to coaching takes the perspective that practice is an opportunity for athletes to make mistakes. Remember that a mistake undetected is a mistake uncorrected. Mistakes tell you the progress your athletes are making in the learning process. Root out errors with patience and enthusiastically help your athletes correct them. Share this perspective with your athletes because they'll appreciate knowing they are permitted to err in their journey to mastering the sport. This will foster a growth mindset within your team where they embrace their mistakes as steps in growing their talent.

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Implementing a positive approach to correcting errors may be easier in individual sports than in team sports. After all, you can focus on each athlete more easily in such sports as wrestling, tennis, track and field, and skating. Furthermore, your comments are usually more private, so they do not embarrass the performer.

Team sports provide unique challenges. How do you provide individual feedback in a group setting during practice using a positive, growth mindset approach? Instead of yelling across the court or field to correct an error (and embarrassing the player), consider substituting for the player who erred. Then you can make the correction individually and reinsert the player into the practice scrimmage. This tactic offers three benefits:

1. Players are more receptive to feedback because they are not being corrected in front of an audience.
2. The other team members remain active, so they don't pay attention to the discussion between you and the erring player. Plus, they are continuing to practice, so they are continuing to develop their technical skills.
3. Since the rest of the team continues to play, you must make corrective comments simple and concise. This saves both you and the player from long, drawn-out explanations that lack value because of their complexity.

In a team setting, you can also use specific positive feedback to emphasize correct group and individual performances. These performances and the accompanying feedback serve to reinforce the kinds of behavior and play you want to see. Use this approach only for positive statements. Save your specific negative feedback for individual discussions.

Providing Effective Feedback

Take this short true-or-false test to raise your awareness about giving feedback:

Questions

1. Save feedback until the end of practice so as not to disrupt practice time.
2. More frequent feedback is better than less frequent feedback.
3. When an athlete is making several technical errors, it is best to correct only one error at a time.
4. You and your assistant coaches should be the only people providing feedback in practice.
5. When giving feedback, do not tell the athlete what was done incorrectly—only provide feedback on how to do the technique correctly.
6. Give simple and precise information about how performance can be improved.
7. Frequent positive feedback, such as "Nice job!", is more important than lots of technical instruction.
8. Use sight and sound in providing feedback.

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Answers

1. *False*—The sooner you give feedback, the more likely athletes will remember what the feedback pertains to and to practice correctly.
2. *True, within reason*—The more often athletes get useful feedback, the more they will try to correct their performance, and thus, the faster their learning will be. As athletes' technical skills improve, though, they need to learn to rely more on their own feedback and less on feedback from the coach.
3. *True*—Learning is more effective when an athlete attempts to correct only one error at a time, which means that you must decide which error to correct first. To do so, begin by determining whether one error is causing another. If it is, have the athlete try to correct that error first because this will eliminate the other error(s). However, if the errors seem to be unrelated, have the athlete correct the error that you think will bring the greatest improvement when remedied. Improvement will likely motivate the athlete to correct the other error(s).
4. *False*—Athletes, especially those who are a little older, should give feedback to each other (of course this feedback must be accurate).
5. *False*—In fact, feedback means to feed back exactly what was done. When athletes perform incorrectly, you should feed them back with what they did wrong. Then explain how to do the technique correctly.
6. *True*—Tell and show your athletes what they must do to correct errors. Be careful not to go overboard; give just enough information so they can concentrate on correcting one error at a time.
7. *False*—Positive feedback is good. However, specific positive feedback is much more valuable. Such feedback specifies what was correct and reinforces those aspects for all the athletes in a group. An example of specific positive feedback is, "Nice follow-through on that shot!" The same principle holds for negative feedback. If a player performs incorrectly, simply saying, "That was a terrible shot," is not helpful. The player already knows the shot was terrible. What is important is how to improve. A more effective approach would be to say, "Your shot was off the mark because you allowed your elbow to swing to the outside. Try keeping the elbow tucked in to your side."
8. *True*—People learn in different ways; some gain most from explanations of how to improve, whereas others need demonstrations. Both explanations and demonstrations should incorporate specific feedback. For example, you might demonstrate how a player performed a skill, explain what was good and what you believe needs more attention for improved performance, and demonstrate the refinement you would like to see. Show and tell your players how they can improve using specific positive and negative feedback.

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